EL NIÑO ACTION BRIEFING · 26 SEPTEMBER 2026 · VERIFIED RUN · COFFEE-INDUSTRY ACTIONS FIRST

Ocotal, Nicaragua

Nothing in Nicaragua pays out on a drought trigger. The state has closed most of the channels outside money once used. The one foreign relationship that still reaches the farm gate without permission is the coffee contract. In Nueva Segovia this winter, the harvest is the relief operation.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
The Climate Impact Lab region containing Ocotal, Nueva Segovia	No figure published for Nicaragua or Central America	Jun 2026 – Feb 2027 (heat only)	Drought now; hunger at harvest; dengue from stored water; flowering in 2027; flood after

15/15 recommendations verified. 8 corrected (partially confirmed), 0 confirmed outright, 3 unverified, 4 contraindicated. The funding target moved from $440k to $320k, down 27%, under a third, so the envelope stands. **Ranking rule for this run:** the reader asked to prioritise actions the coffee industry can perform or benefit from, so those come first wherever the evidence allows. **Political caveat:** more than 5,500 organisations, about 80% of those operating, have been shut since 2018. The Nicaraguan Red Cross was dissolved in 2023, and more than 200 clergy have been exiled or barred. Every action here has been filtered for what can be done without endangering the people it reaches. **Method caveat:** the session's web-search budget was exhausted; agents worked by direct fetch.

EXECUTIVE SUMMARY — AT A GLANCE



13.63°N 86.47°W, Nicaragua · Natural Earth

- **The risk.** Drought now, hunger at harvest, and dengue from stored water; after the record 1997–98 El Niño, La Niña's Mitch brought great destruction to Ocotal and killed at least 3,800 nationally.
- **First action.** Importers and roasters keep buying Nueva Segovia coffee at full volume and a fair price if the crop comes in short, and pay early, with contract terms by 15 November.
- **The money.** $150k floor, $320k target, about $445k ceiling with buyer matching; therapeutic food through UNICEF, $60k, is the best value per dollar.
- **Do not.** Do not send money labelled as a donation, or through NGOs, churches, or the Red Cross: over 5,500 organisations are shut, and donation-shaped money exposes its recipients.

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* Suggested formulation: check with a health worker before sharing.

The national weather service, the health ministry, and the main Nicaraguan newspapers were unreachable, and no Nicaragua-specific 2026 rainfall forecast was found in any source.

EXPOSURE, HONESTLY

WHAT IS KNOWN AND WHEN IT BITES

- **Where this is.** Ocotal has about 50,000 people at 500 to 610 metres and is the capital of Nueva Segovia (275,291 people, 2021 estimate). It sits in the Dry Corridor, among the pine and coffee of Dipilto, Mozonte, and Macuelizo, near the Honduran border at Las Manos.
- **This El Niño.** Niño-3.4 was 1.8 °C above normal in August. The US Climate Prediction Center gives more than 90% odds of a very strong event and 75% that October–December sets a record since 1950.
- **The timeline that matters.** The Central American climate forum (July 2026) reports a late start to the rains and an early mid-summer drought from late June, with effects "already being felt." Next door in Honduras, the forecast showed August deficits near 50% in the Dry Corridor. The forum published no forecast for Nicaragua. So: **now to October**, drought damages the second planting, and losses show at the November–December harvest. **November to March**, El Niño peaks in the dry season and during the coffee harvest, and water and food run short. **April to June 2027**, a late or erratic start to the rains threatens the next planting and, by assumption, the coffee flowering. **Late 2027**, if La Niña follows, hurricane risk rises: Mitch in 1998 struck in La Niña after a record El Niño.
- **Water was short before the drought.** In February 2026 the national water utility's president named Ocotal among cities with a supply deficit.
- **Who is at risk.** National child stunting was 17.3% in 2012, the latest survey, down from 30.5% in 1998; no figure for Nueva Segovia exists. In 2019 Nicaragua had the highest dengue incidence in the Americas, 2,962 suspected cases per 100,000 (PAHO); no national figure since. Smallholders grow 79% of the country's basic grains. Remittances are 26.6% of GDP, and harvest labour costs doubled in 2024/25 as workers left.
- **The forest.** A local environmental group estimates the Dipilto–Jalapa pine fell from 43,000 to 17,000 hectares through logging and bark beetle; about 300,000 people would be affected if it disappears.

WHAT THIS PLACE SHOULD DO

COFFEE INDUSTRY FIRST

The first six actions have a coffee-industry owner or payer. They are ranked by lives protected per unit of effort, and by how safely money moves through them. Six of seven lenses reached the same core idea independently: from November to February, the harvest wage is the only cash

* Suggested formulation: check with a health worker before sharing.

that reliably reaches hungry households here.

Buyers: keep buying Nueva Segovia coffee at full volume and a fair price if the crop comes in short, and pay the harvest early

PARTIALLY CONFIRMED

SKEPTIC · EMERGENCY MGR · ECONOMIST · PHILANTHROPIST · FRONTLINE

ADAPTATION INVENTORY: PRECEDENT, MEXICO PROGRESA OFFSET DROUGHT'S EFFECT ON DIET

OWNER	BY WHEN	COST	PREVENTS
Importers and roasters (e.g., Cafe Imports, which buys from PRODECOOP and Dipilto micromills)	Contract terms by 15 November	About a quarter's working-capital interest on early payment (estimate)	Hunger in the barrios before any official figure shows it

The verified ranking of safe channels puts a buyer paying a certified cooperative for coffee first, ahead of UN agencies working through ministries. No coffee cooperative has been reported cancelled. Cutting volume or delaying payment when the crop is short is the single most damaging thing a buyer can do here, because the harvest wage is the region's biggest cash transfer. Precedent: during the 2001 coffee crisis, Nicaragua's own cash-transfer programme protected household spending most for families hit hardest (Maluccio, IFPRI).

* Suggested formulation: check with a health worker before sharing.

Legal caution, verified in part: summaries of the foreign-agents law (Ley 1040) and the non-profit law (Ley 1115) aim at "foreign funds" and donations, not export sales. But Ley 1040 reportedly counts funding from foreign companies in its definition, and in March 2025 a credit cooperative and a cooperative federation were evicted in a government raid. Keep every payment tied to invoiced coffee, never call it a grant or donation, and get a Nicaraguan legal opinion before paying advances.

Lenders and buyers: write a drought clause into the 2026/27 pre-harvest credit already being negotiated, deferring repayment automatically if the 2027 flowering fails

PARTIALLY CONFIRMED · ECONOMIST · SKEPTIC · VERIFICATION

ADAPTATION INVENTORY: PRECEDENT, US CONTINGENT LOANS CUT POST-DISASTER CLOSURES

OWNER	BY WHEN	COST	TRIGGER
Cooperative lenders and importers financing PRODECOOP-type cooperatives	Before contracts close, by 15 November	No new product; contract wording	Measured flowering or cherry-weight loss, Feb–Apr 2027

This is the cheapest risk transfer available before the peak, and it needs nothing to be created. Nothing else in Nicaragua pays on a drought. The regional catastrophe pool, CCRIF, covers Nicaragua for cyclone, earthquake, and excess rain; it has paid about $41.2 million since 2016, including $19.9 million after Iota, but it has no drought product. The UN's Dry Corridor drought framework dropped Nicaragua in 2025; OCHA's own monitoring code says it is watched "informally." Pre-harvest finance repaid in coffee is standard; the lenders often named (Root Capital, Shared Interest, Oikocredit) could not be confirmed this run. The Skeptic's warning shapes the design: a blanket rescue loan before losses are measured turns one bad season into years of debt, so the clause defers rather than lends.

Cooperatives: set aside part of the Fairtrade Premium as a drought reserve, by committee resolution

PARTIALLY CONFIRMED · ECONOMIST · HISTORIAN · VERIFICATION · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
PRODECOOP's premium committee and similar certified cooperatives	Next committee meeting	Reallocation of existing premium	Members facing the 2027 flowering with no reserve

* Suggested formulation: check with a health worker before sharing.

PRODECOOP is confirmed: founded 1993, Fairtrade, organic, and small-producer certified, with collection centres in Dipilto, Jalapa, Quilalí, San Juan del Río Coco, and Palacagüina, exporting to Europe, North America, and Asia. The Fairtrade Premium is already a cooperative-controlled fund, and cooperatives already spend it on drought-resistant seedlings. It is not escrowed and not event-triggered today. The premium rate usually quoted ($0.20/lb) could not be confirmed this run.

Estates: pay pickers weekly and in full, and pilot a harvest-season top-up for picker households with young children through named Dipilto estates

PARTIALLY CONFIRMED · REDESIGNED | PHILANTHROPIST · EMERGENCY MGR · ECONOMIST · PUBLIC HEALTH · HISTORIAN

ADAPTATION INVENTORY: PRECEDENT, NICARAGUA CASH TRANSFER PLUS PRODUCTIVE GRANT

OWNER	BY WHEN	COST	PREVENTS
Dipilto estates and micromills (e.g., Cafetos de Segovia), funded through buyer contract premiums	Harvest, November to February	$120k pilot: 750 households × $40 × 4 months	Children going hungry when grain stocks run out

Redesigned in verification. **PRODECOOP pays its member farmers, not hired pickers, so it has no picker payroll.** Only estates keep picker records, and none in Nueva Segovia could be confirmed. One estate group named by a lens has no farms in Nueva Segovia and was removed. $40 a month is about 26% of the agricultural minimum wage (C$5,721 from March 2024, at C$36.62 per dollar), which is

* Suggested formulation: check with a health worker before sharing.

meaningful. Evidence: Nicaragua's Red de Protección Social cut stunting by 5.3 percentage points in neighbouring Madriz and Matagalpa, significant only at the 10% level. The Adaptation Inventory's Nicaragua entry found that a cash transfer with a productive grant improved food security and assets against weather shocks.

Estates and cooperatives: safe water and container clean-up in harvest camps and picker housing, run like a health-ministry jornada

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST · EMERGENCY MGR | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Farm owners, cooperative members, dry mills	Before pickers arrive in November	$90k for 60 camps (unit cost unsourced) + $50k container clean-up	Diarrhoea and dengue where drought, storage, and crowding meet

The Public Health lens's mechanism: drought makes people store water, stored water breeds Aedes, and the harvest packs children into camps with water drums at the peak. The Camino Verde cluster trial cut dengue infection by 29.5% (3.8–55.3), pooled across Nicaragua and Mexico. Household chlorination cut diarrhoea about a quarter, filtration about half (Cochrane, low and moderate certainty). The only study of picker housing found poor housing, crowding, and inadequate toilets among Nicaraguan pickers working in Costa Rica; none exists for Nicaragua itself. Frame it as farm hygiene and quality compliance, not as a programme.*

Estates: screen picker children's arm circumference at the camps and refer severe cases, with therapeutic food supplied through UNICEF to the health ministry

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Estates for screening; UNICEF and the health ministry for treatment	From the start of harvest	$60k therapeutic food and micronutrients	Severe acute malnutrition going untreated and uncounted

* Suggested formulation: check with a health worker before sharing.

Community treatment of severe acute malnutrition costs about $26 per healthy life-year gained in Bangladesh (Puett 2013) and $53 in a Zambian model (Bachmann 2009), the best value per dollar in this report. UNICEF operates under a 2024–28 agreement with the government covering nutrition. The state controls the channel, so the donor cannot target Nueva Segovia; with no official malnutrition figure for the department, the camps may be the only place these children are counted.*

Wet mills: agree water draw with the local water committees and fill storage before the dry season

UNVERIFIED | EMERGENCY MGR · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Wet and dry mills; community water committees (CAPS)	31 December	$5–20k per mill for low-water pulpers (estimate)	Coffee processing competing with drinking water from February

The water deficit in Ocotal is confirmed; no source for wet-mill water use or savings from low-water equipment could be found. The reasoning is sound and the mills gain from it; the numbers are not.

Fire breaks and beetle watch on the coffee–pine edge before the March–May fire season

UNVERIFIED | EMERGENCY MGR · HISTORIAN · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Cooperatives and farm owners with the forestry institute and municipalities	31 January, using post-harvest labour	$10–30k per municipality (estimate)	Losing the shade pine and watershed that coffee depends on

* Suggested formulation: check with a health worker before sharing.

The pine loss is sourced to a local group's estimate. The claim that bark beetle outbreaks followed past El Niño droughts here, 1999–2002 in Nicaragua and 2014–17 in Honduras, could not be sourced with hectares or dates; a Honduran study links beetle susceptibility to drought in 2017–19. Coffee has the interest: the shade and the water come from that pine.

Plan now for late 2027: keep homes and wet mills out of the Dipilto floodplain, and renew CCRIF excess-rainfall cover

PARTIALLY CONFIRMED · HISTORIAN · ECONOMIST · EMERGENCY MGR

ADAPTATION INVENTORY: PRECEDENT, NICARAGUA BRIDGES OFFSET FLOOD INCOME LOSS

OWNER	BY WHEN	COST	PREVENTS
Mills and farm owners; municipalities; the Ministry of Finance for CCRIF	Before the 2027 hurricane season	CCRIF premium (unpriced)	A Mitch-type flood on ground stripped by drought and fire

Mitch left Ocotal with "great destruction," and streams near Dipilto rose more than 10 metres above normal; the Pueblos Unidos barrio was built afterwards to resettle families. Mitch (1998) and Eta and Iota (2020) both struck in La Niña; Mitch followed the record 1997–98 El Niño. The Adaptation Inventory's Nicaragua entry found that permanent bridges fully offset the 18% labour-income loss from flood-cut roads. Correction: Iota's landslide deaths at Wiwilí were in Jinotega department, not Nueva Segovia.

Ocotal: publish a firm water-rationing schedule, and hand out lids for stored water

PARTIALLY CONFIRMED · EMERGENCY MGR · PUBLIC HEALTH · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Municipality and the water utility; health ministry	15 December	Low tens of thousands of dollars (estimate), municipal budget	Disorderly shortages, and the dengue that storage breeds

* Suggested formulation: check with a health worker before sharing.

The supply deficit is confirmed. The widely quoted report of cuts from 5:30 am to 9 pm came from one social-media user and was not tied to Ocotal. State bodies own this action; it is included because nothing else protects the town's barrios.

After this season: design rainfall-index cover for coffee flowering, paid for by exporters and roasters

UNVERIFIED · NOT BEFORE 2027–28 | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	TRIGGER
Exporters, roasters, an insurer	Design in 2027	Unpriced	February–April rainfall below a set percentile

No such product exists in Nicaragua, and no weather station at Dipilto or Mozonte could be confirmed; one is believed to exist at Ocotal. This is new structuring work.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Importers & roasters (e.g., Cafe Imports)	Exporters	Cooperative lenders	PRODECOOP & cooperatives	Dipilto estates & farm owners	Wet & dry mills	Water committees (CAPS)	Municipalities	Water utility	Health ministry	UNICEF	Forestry institute	Ministry of Finance	CCRIF & insurers
1. Keep buying at full volume, pay early	**A/R**			**I**	**I**									
2. Drought deferral clause in pre-harvest credit	**R**		**A/R**	**C**										
3. Fairtrade Premium drought reserve				**A/R**										
4. Weekly picker pay and child top-up pilot	**R**				**A/R**									
5. Safe water and container clean-up in camps	**R**			**R**	**A/R**	**R**								
6. Screen picker children, refer for UNICEF treatment					**A/R**					**R**	**R**			
7. Mills agree water draw with CAPS						**A/R**	**R**							
8. Fire breaks and beetle watch, pine edge				**A/R**	**R**			**C**				**R**		
9. Floodplain siting and CCRIF excess-rainfall cover					**R**	**R**		**A**					**R**	**C**
10. Ocotal water-rationing schedule and storage lids								**A/R**	**R**	**R**				
11. Rainfall-index cover for flowering, from 2027	**R**	**A/R**												**R**

What the chart shows. **Estates are accountable for three rows, and no estate in Nueva Segovia with picker records could be confirmed.** Rows 4, 5, and 6 all depend on named Dipilto estates that keep picker lists, which is the funding section's largest uncertainty. **Importers and roasters appear on five rows but are accountable for only one.** Their main role is paying premiums that others spend, which keeps the money in the form of invoiced coffee. **State bodies own the town's own protection.** Rows 9 and 10 sit with the municipality, the water utility, the health ministry, and the Ministry of Finance, and the treatment half of row 6 runs through UNICEF and the health ministry, which control targeting. **Row 11 names three owners and no lead.** The A is placed on exporters as a judgement call, and the product does not yet exist in Nicaragua.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled and verified. The safest money in Ocotal is not a donation; it is paid as coffee. Buyer-paid actions 1 to 3 cost little or nothing and sit outside the envelope; the envelope funds what they cannot.

* Suggested formulation: check with a health worker before sharing.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$150k	$320k	About $445k with buyer matching	Whether any estate keeps picker lists that can deliver a household top-up

Arithmetic. Floor: camp water $90k + therapeutic food through UNICEF $60k = $150k. Target: floor + picker top-up pilot 750 × $40 × 4 = $120k + container clean-up $50k = $320k. Ceiling: target + a buyer match of about $125k, from a $0.10/lb premium on 30 containers (275 bags × 69 kg = 41,833 lb, so $4,183 per container) = about $445k. The volume of 30 containers is unverified. The lens proposed $440k; the target fell 27%, under a third, because the cooperative has no picker payroll and the household count was unsourced.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Farm owners, cooperative members, dry mills	Action 5: water treatment and safe storage in 60 harvest camps	$90k	Paid through buyer contract premium to named estates	Cochrane household water treatment; $1,500 per camp unsourced. Conditional on naming the estates that run camps.	Keep, conditional
UNICEF, for the health ministry in Nueva Segovia	Action 6: therapeutic food and micronutrients	$60k	Grant to UNICEF	$26–53 per healthy life-year (Puett 2013; Bachmann 2009). The only placement with no coffee partner; state controls targeting.	Keep
Named Dipilto estates and micromills, e.g., Cafetos de Segovia; premium written into importer contracts	Action 4: harvest top-up for 750 picker households with young children	$120k	Contract premium plus funder grant paid to the importer	$40 is ~26% of the agricultural minimum wage. Halved from 1,500 households: PRODECOOP has no picker payroll, and no count of picker households exists.	Reduced, pilot
Cooperative members	Action 5: container clean-up against dengue	$50k	Via cooperatives, as health-ministry-style jornadas	Camino Verde, −29.5% dengue infection (Nicaragua and Mexico). Cost unsourced.	Keep, low confidence

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Importers (e.g., Cafe Imports)	Buyer match on the picker top-up	~$125k	$0.10/lb premium on ~30 containers	Arithmetic confirmed; volume unverified.	Ceiling only

The funder's one thing survives, and verification sharpened it: pay it as coffee. A roaster's premium arrives on the payroll at harvest, right at the peak, and does not mark the family as a recipient of foreign money. But label it, invoice it, and contract it as coffee, and get the legal opinion first. The window to put it into this season's contracts closes in November.

DO NOT DO THIS — CONTRAINDICATED

Do not send money labelled as a donation, or through NGOs, churches, or the Red Cross

CONTRAINDICATED · SKEPTIC · PHILANTHROPIST · FRONTLINE; CONFIRMED

More than 5,500 organisations, about 80%, have been shut since 2018. The Nicaraguan Red Cross was dissolved on 9 May 2023 and its property seized. Bishop Rolando Álvarez, who led the diocese covering this region, was sentenced to 26 years and expelled in January 2024 with 15 priests. Even the Green Climate Fund terminated its Bio-CLIMA project in 2024. Money shaped like a donation exposes the people who receive it.

* Suggested formulation: check with a health worker before sharing.

Instead: invoiced coffee payments, contract premiums, and UN agencies working through ministries, in that order. As the Frontline lens put it: when nothing carries an outside name, nothing gets shut down.

Do not cut volume or delay payment because the crop is short

CONTRAINDICATED SKEPTIC · EMERGENCY MGR · ECONOMIST

The harvest wage is the only cash that reliably reaches hungry households from November to February. A buyer who leaves in the drought year removes the relief operation, and will be absent when the floods follow.

Do not offer blanket rescue credit before losses are measured

CONTRAINDICATED SKEPTIC

A loan against a one-season loss becomes debt carried for years; the 2001 coffee crisis closed farms that borrowed through it.

Instead: have dry mills weigh cherry and measure density in November and December, then give harvest-contingent advances or payment extensions only to farms with measured losses, and write the deferral clause in action 2.

Do not wait for the UN drought framework to trigger

CONTRAINDICATED VERIFICATION

OCHA removed Nicaragua from its Dry Corridor anticipatory-action framework in 2025 and now monitors it only informally. No Nicaragua allocation triggered in 2024 to 2026. FEWS NET's Nicaragua coverage stopped in October 2024. There is no Red Cross emergency fund channel since 2022. Nothing will arrive on a forecast.

* Suggested formulation: check with a health worker before sharing.

DATE	EVENT	TOLL	WHAT FOLLOWED
1997–98	Record El Niño; Dry Corridor drought	—	La Niña; Mitch
Oct 1998	Hurricane Mitch, in La Niña: Ocotal "great destruction"; streams near Dipilto rose 10+ m	At least 3,800 dead nationally	Pueblos Unidos barrio built to resettle families (with Basque NGO HONEK)
2000–06	Red de Protección Social in Madriz and Matagalpa	—	Stunting –5.3 points; protected spending in the 2001 coffee crisis
2001	Coffee price collapse	Farms closed across the north	—
2009, 2014–16	El Niño droughts in the Dry Corridor	2009: 50–100% of crops affected; 2014–16: millions needing food aid	—
2019	Dengue epidemic	2,962 suspected cases per 100,000; 30 deaths nationally	Highest rate in the Americas
Nov 2020	Eta and Iota, in La Niña	Iota: 28 dead, 29 missing; 63,000 evacuated; 47,638 families lost water	CCRIF paid ~$30.6M for Eta and Iota
2018–2025	More than 5,500 organisations shut; Red Cross dissolved (2023); bishop of Estelí expelled (2024)	—	Donation channels largely closed
2025	OCHA removes Nicaragua from its drought framework	—	No trigger-based funding
Feb 2026	Water utility admits supply deficit in Ocotal	—	—
Jun–Aug 2026	Late rains and early mid-summer drought reported across the region	Losses due at the Nov–Dec harvest	This briefing

* Suggested formulation: check with a health worker before sharing.

The recurrence nobody has noticed, from the Historian: El Niño itself has not flooded Ocotal. What came after it did. Drought in 1997–98 was followed by Mitch, then forest loss, then the 2001 price collapse; the damage stacked over about four years. Buyers who leave in the drought year are gone when the floods come.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Route money through cooperatives** versus **contract it to named farms**. Three lenses favour cooperatives; the Skeptic warns they can be absorbed or closed. Verification found no coffee cooperative cancelled but did find a credit cooperative evicted in 2025, and found that the main cooperative has no picker payroll. Resolution: pay the cooperative for coffee, pay pickers through estates, and keep everything invoiced.
- **The damage is done** versus **the worst is 2027–28**. Both hold, at different times: the drought is happening now, the losses show at harvest, the flowering risk is spring 2027, and the flood risk is late 2027.
- **Does El Niño hit mountain coffee?** A 2025 study of Latin America found El Niño exposure everywhere but larger productivity effects in lowland cacao than mountain coffee, using satellite productivity rather than yields. The coffee signal is real but not certain.

WHAT NOBODY COVERED — BLIND SPOT

No lens counted the pickers. There is no figure for how many work the Nueva Segovia harvest, how many bring children, or how many cross from Honduras at Las Manos, and the funding placements rest on that number. No lens looked at the grandparents left raising children while parents work abroad, whose only lifeline is a remittance that a US labour-market shock could cut. And no lens could reach Nicaragua's weather service, health ministry, or main newspapers, so the local picture of this year's rains and this year's hunger is inferred from the region around it.

HOW TO READ THIS — METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

* Suggested formulation: check with a health worker before sharing.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

This run. At the reader's request, actions that the coffee industry can perform or benefit from were ranked first, and all seven lenses were asked to name coffee-industry owners and payers. Every action was also filtered for political safety. The session's web-search budget was exhausted, so all thirteen agents worked by direct fetch; OCHA's position came from its public code repositories. Six verifiers checked fifteen recommendations; the verdict tags carry the result.

The El Niño projection covers *heat only* and publishes nothing for Central America. The Adaptation Inventory was consulted through its data feed (96 entries, 30 countries): it holds two Nicaragua entries, both cited, and nothing on coffee, dengue, or harvest labour.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Ocotal, Nicaragua. Climate Impact Lab impact region, as the report records it: The Climate Impact Lab region containing Ocotal, Nueva Segovia. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-

* Suggested formulation: check with a health worker before sharing.

city detail is used for hazard exposure and action design, not for death counts.

- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (safe stored water and dengue in harvest camps, and screening and treating child malnutrition) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

* Suggested formulation: check with a health worker before sharing.

References (verification status in brackets) · CPC ENSO discussion cpc.ncep.noaa.gov · ONI table cpc.ncep.noaa.gov · Central American climate forum LXXXI recursoshidricos.org · FEWS NET El Niño 2026–27 fews.net · FEWS NET Nicaragua fews.net [confirmed; no Nicaragua forecast] · González-González et al. 2025 doi.org · Central American dry corridor wikipedia [partially confirmed] · Ocotal wikipedia · es.wikipedia · Dipilto es.wikipedia · Nueva Segovia wikipedia · pine loss confidencial.digital · water deficit confidencial.digital · Hurricane Mitch wikipedia · Hurricane Iota wikipedia · Hurricane Eta wikipedia [confirmed with corrections] · HRW World Report 2026, Nicaragua hrw.org · Nicaraguan Red Cross wikipedia · Bishop Álvarez wikipedia · GCF Bio-CLIMA greenclimate.fund · ICNL civic freedom monitor icnl.org [confirmed; legal text not read] · PRODECOOP prodecoop.com · Cafe Imports and PRODECOOP cafeimports.com · origin report 2018 cafeimports.com · Fincas Mierisch farms fincasmierisch.com [removed: no Nueva Segovia farms] · WFP Nicaragua wfp.org · UNICEF Nicaragua unicef.org · exchange rate bcn.gob.ni · minimum wage wikipedia · CCRIF payouts ccrif.org · World Bank projects worldbank.org · OCHA Dry Corridor monitoring github.com/OCHA-DAP · OCHA framework register github.com/OCHA-DAP · IFRC appeals ifrc.org [confirmed] · Stunting, World Bank api.worldbank.org · PAHO dengue 2019 paho.org · Camino Verde PMC4495677 · Puett 2013 doi.org · Bachmann 2009 PMC2630929 · Red de Protección Social cgspace · coffee crisis role repec.org · picker housing, Costa Rica doi.org [confirmed with corrections] · Demoted or unverified: bark-beetle hectares for 1999–2002 and 2014–17; Quesungual outperforming through Mitch; wet-mill water savings; the named pre-harvest lenders; the Fairtrade premium rate; a Dipilto or Mozonte weather station; Nicaragua's first-to-join CCRIF status; OCHA framework dollar amounts; picker counts; coffee flowering with the first rains.
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions; Nicaragua cash transfer and productive grant entry; Nicaragua bridges entry; US contingent loans entry)
Generated 26 September 2026 for Ocotal, Nueva Segovia, Nicaragua. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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